

tsvc/s241 · s241.c · bare_llm · claude-haiku-4-5-20251001 · BROKEN

```
-0 / +10 lines of 176 . 2 parallel region(s) added . 0 of 0 candidate(s) kept . RESTRUCTURING (2 code line(s) added, 0 removed) . restructuring level 0
```

BEFORE

132 }

133

```
134 static real t_kernel s241(void)
```

135 {

```
136     for (int nl = 0; nl < R; nl++) {
```

```
137     for (int i = 0; i < LEN 1D-1; i++) {
```

```
138     a[i] = b[i] * c[i] * d[i];
```

```
139         b[i] = a[i] * a[i+1] * d[i];
```

140 }

```
141 pb mix(nl);
```

142

```
143     return (real t)0;
```

144 }

145

AFTER

132

133

```
134 static real t_kernel_s241(void
```

135 +

```
136     for (int nl = 0; nl < R; nl++) {
```

```
137         // Phase 1: compute all a[i] values in parallel
```

```
139         #pragma omp parallel for
```

```
140     for (int i = 0; i < LEN_1D-1; i++) {
```

```
141         a[i] = b[i] * c[i] * d[i]
```

```
144 // Phase 2: compute all b[i] values in parallel using updated a values
```

```
145 // Each iteration i writes b[i], reads a[i], a[i+1], d[i] – no loop-carried
```

```
146      #pragma omp parallel for
```

```
147     for (int i = 0; i < LEN_1D-1; i++) {
```

```
148         b[i] = a[i] * a[i+1] * d[i]
```

149

156

```
151 pb mix(nl);
```

152

```
153     return (real t)0;
```

154

155

file